

Random Matrices

M1R Project

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Overview

I Central Limit Theorem: Before digging in to random matrices we discuss a baby-version of universality coming from adding many random variables, using the Method of Moments to relate to the number of partitions of a set into pairs.

II Universality: Many families of random matrices have the same eigenvalue statistics which can be explained by adapting the Method of Moments to relate to the number of *non-crossing* partitions of a set into pairs.

III Free Probability: Adding or multiplying random matrices satisfy special rules of (non-commutative) free random variables. These can be explained again by the Method of Moments.

IV Randomised Linear Algebra: We finish with a nice application of using randomness to reduce linear algebra problems like linear regression to lower dimensional versions which are faster to solve.



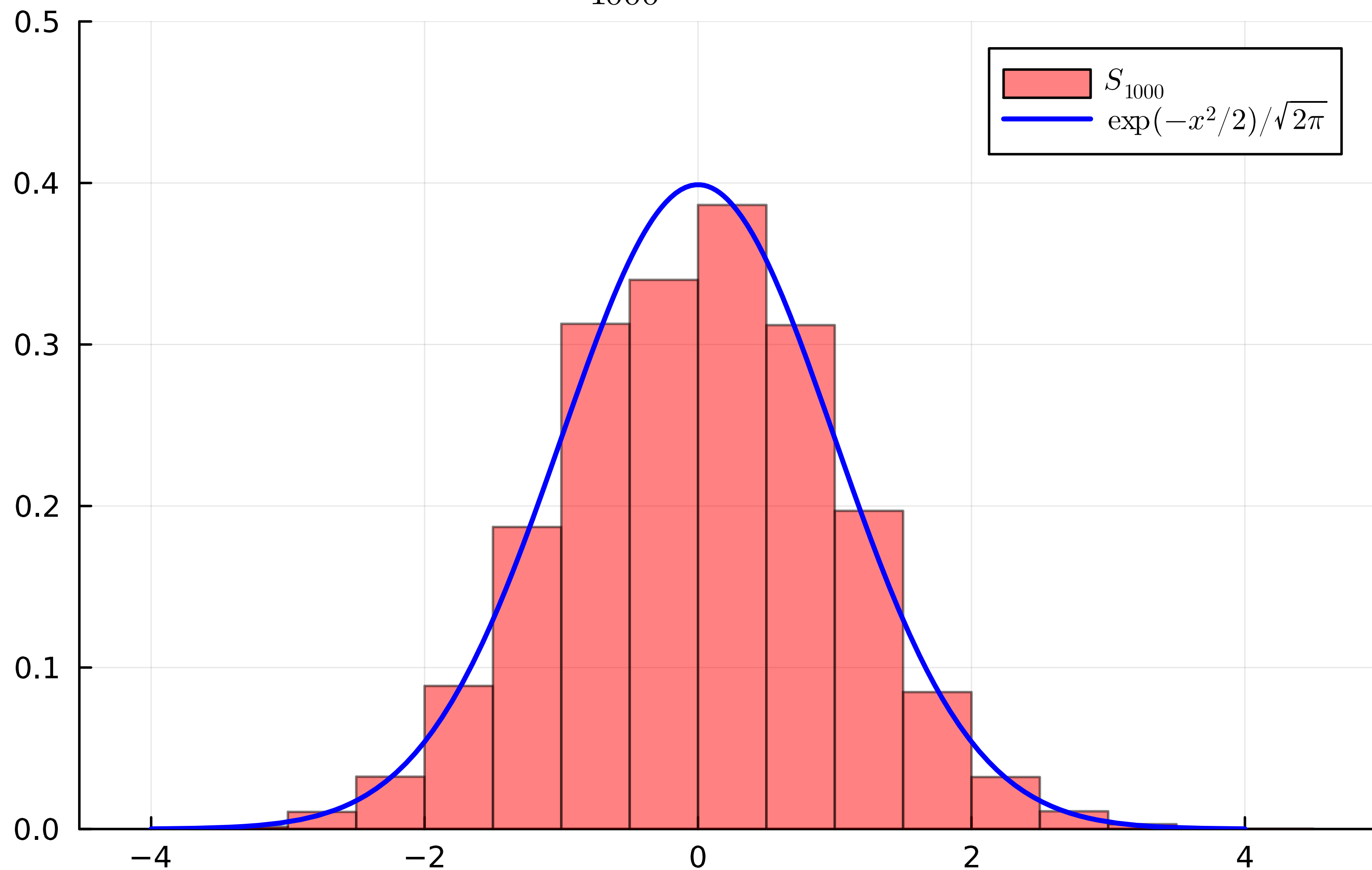
Link to notes

I Central Limit Theorem

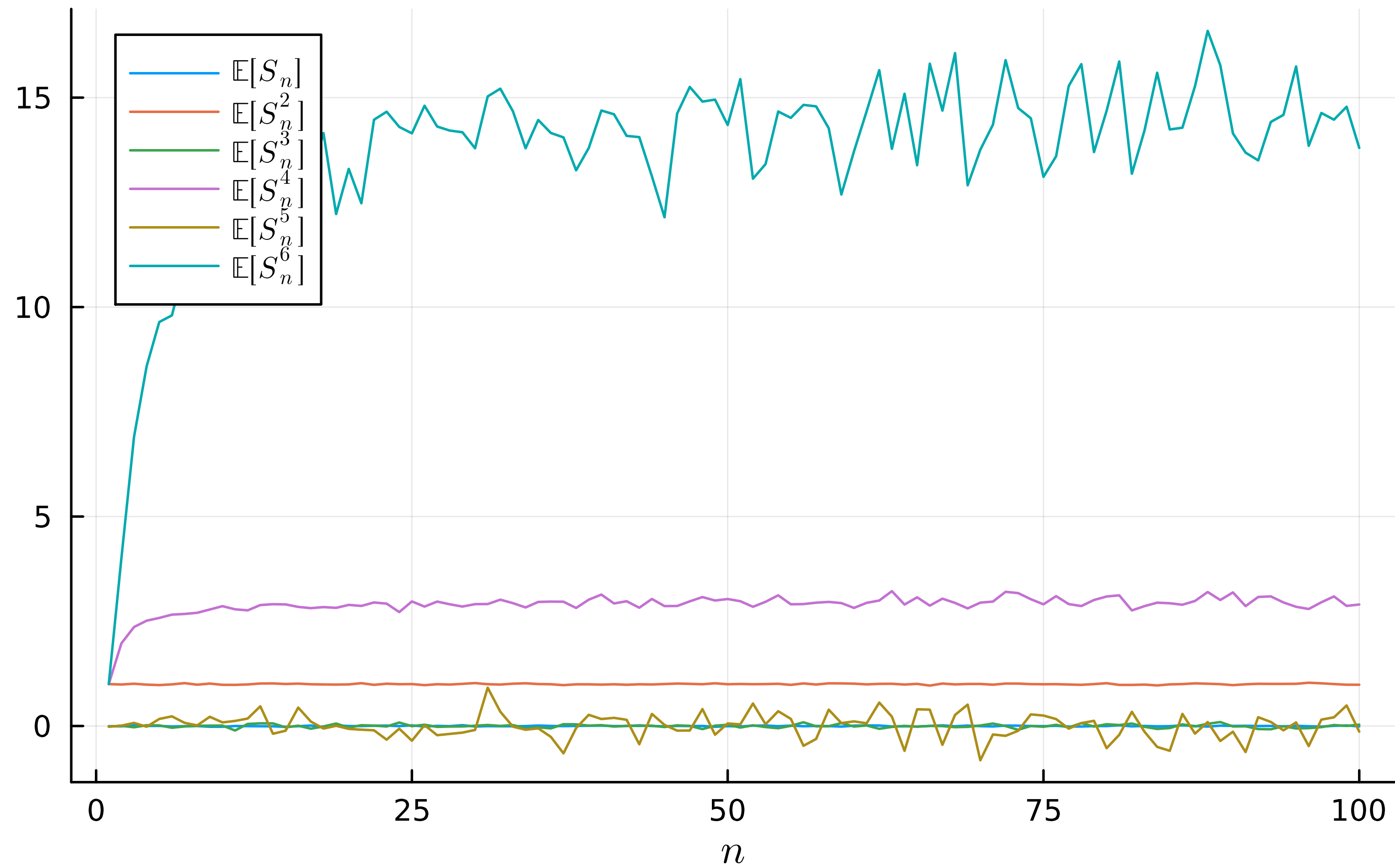
I.1 Experimental observation of the CLT

Sums of random variables look like Gaussians

Histogram of S_{1000} , sampled 10000 times



Moments of S_n



I.2 Moments of the normal distribution

Expressible as a double factorial

Find a formula for $\mu_k^N := \int_{-\infty}^{\infty} x^k \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx.$

Definition 1 (double factorial).

$$(2m-1)!! := \frac{(2m)!}{2^m m!}$$

Pairings of 1,...,6:

$\{1, 2\}, \{3, 4\}, \{5, 6\}$	$\{1, 2\}, \{3, 5\}, \{4, 6\}$	$\{1, 2\}, \{3, 6\}, \{4, 5\}$
$\{1, 3\}, \{2, 4\}, \{5, 6\}$	$\{1, 3\}, \{2, 5\}, \{4, 6\}$	$\{1, 3\}, \{2, 6\}, \{4, 5\}$
$\{1, 4\}, \{2, 3\}, \{5, 6\}$	$\{1, 4\}, \{2, 5\}, \{3, 6\}$	$\{1, 4\}, \{2, 6\}, \{3, 5\}$
$\{1, 5\}, \{2, 3\}, \{4, 6\}$	$\{1, 5\}, \{2, 4\}, \{3, 6\}$	$\{1, 5\}, \{2, 6\}, \{3, 4\}$
$\{1, 6\}, \{2, 3\}, \{4, 5\}$	$\{1, 6\}, \{2, 4\}, \{3, 5\}$	$\{1, 6\}, \{2, 5\}, \{3, 4\}.$

Lemma 1 (pair partitions). *The number of partitions of $2m$ into pairs equals $(2m - 1)!!$.*

Lemma 2 (Gaussian moments).

$$\mu_k^N := \int_{-\infty}^{\infty} x^k \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx = \begin{cases} 0 & \text{if } k \text{ is odd} \\ (k-1)!! & \text{if } k \text{ is even} \end{cases}$$

I.3 Proof of CLT using Method of Moments

Relate moments of sums to partitions

Theorem 1 (CLT). *Let $X_1, X_2, \dots$ be iid variables with $\mathbb{E}[X_j] = 0$, $\mathbb{E}[X_j^2] = 1$, and all other moments are finite. Then for all k we have*

$$\mathbb{E}[S_n^k] \xrightarrow{n \rightarrow \infty} \mu_k^N.$$

Partition	Term	Count
$\{1\}, \{2\}, \{3\}$	$\mathbb{E}[X_{i_1} X_{i_2} X_{i_3}] = \mu_1^3 = 0$	$n(n-1)(n-2) = n^3 + O(n^2)$
$\{1, 2\}, \{3\}$	$\mathbb{E}[X_{i_1}^2 X_{i_3}] = \mu_2 \mu_1 = 0$	$n(n-1)$
$\{1, 3\}, \{2\}$	$\mathbb{E}[X_{i_1}^2 X_{i_2}] = \mu_2 \mu_1 = 0$	$n(n-1)$
$\{1\}, \{2, 3\}$	$\mathbb{E}[X_{i_1} X_{i_2}^2] = \mu_1 \mu_2 = 0$	$n(n-1)$
$\{1, 2, 3\}$	$\mathbb{E} X_{i_1}^3 = \mu_3$	n

Partition	Term	Count
$\{1\}, \{2\}, \{3\}, \{4\}$	$\mathbb{E}[X_{i_1}X_{i_2}X_{i_3}X_{i_4}] = \mu_1^4 = 0$	$n(n-1)(n-2)(n-3) = n^4 + O(n^2)$
$\{1, 2\}, \{3\}, \{4\}$	$\mathbb{E}[X_{i_1}^2X_{i_3}X_{i_4}] = \mu_2\mu_1^2 = 0$	$n(n-1)(n-2) = n^3 + O(n^2)$
$\{1, 3\}, \{2\}, \{4\}$	$\mathbb{E}[X_{i_1}^2X_{i_2}X_{i_4}] = \mu_2\mu_1^2 = 0$	$n(n-1)(n-2) = n^3 + O(n^2)$
$\{1, 4\}, \{2\}, \{3\}$	$\mathbb{E}[X_{i_1}^2X_{i_2}X_{i_3}] = \mu_2\mu_1^2 = 0$	$n(n-1)(n-2) = n^3 + O(n^2)$
$\{1\}, \{2, 3\}, \{4\}$	$\mathbb{E}[X_{i_1}X_{i_2}^2X_{i_4}] = \mu_1\mu_2\mu_1 = 0$	$n(n-1)(n-2) = n^3 + O(n^2)$
$\{1\}, \{2, 4\}, \{3\}$	$\mathbb{E}[X_{i_1}X_{i_2}^2X_{i_3}] = \mu_1\mu_2\mu_1 = 0$	$n(n-1)(n-2) = n^3 + O(n^2)$
$\{1\}, \{2\}, \{3, 4\}$	$\mathbb{E}[X_{i_1}X_{i_2}X_{i_3}^2] = \mu_1^2\mu_2 = 0$	$n(n-1)(n-2) = n^3 + O(n^2)$
$\{1, 2\}, \{3, 4\}$	$\mathbb{E}[X_{i_1}^2X_{i_3}^2] = \mu_2^2 = 1$	$n(n-1) = n^2 + O(n)$
$\{1, 3\}, \{2, 4\}$	$\mathbb{E}[X_{i_1}^2X_{i_2}^2] = \mu_2^2 = 1$	$n(n-1) = n^2 + O(n)$
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$\{1, 2, 3\}, \{4\}$	$\mathbb{E}[X_{i_1}^3X_{i_4}] = \mu_3\mu_1 = 0$	$n(n-1) = n^2 + O(n)$
$\{1, 2, 4\}, \{3\}$	$\mathbb{E}[X_{i_1}^3X_{i_3}] = \mu_3\mu_1 = 0$	$n(n-1) = n^2 + O(n)$
$\{1\}, \{2, 3, 4\}$	$\mathbb{E}[X_{i_1}X_{i_2}^3] = \mu_1\mu_3 = 0$	$n(n-1) = n^2 + O(n)$
$\{1, 2, 3, 4\}$	$\mathbb{E}[X_{i_1}^4] = \mu_4$	n

I.4 Poster Projects

Ideas for posters related to CLT

Comparison with Moment Generating Function proof of CLT

Explore multiplicative versions of CLT

Explore heavy-tail versions of CLT